Solutions of the KP_real NCq3 chain

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Contents

1 KP_real **3**

A következőképp kell olvasni az eredményeket:

1. h_tilde: Az első egy olyan klasszifikáció, ahol nem volt az extra $\tilde{h}$ tag a $\mathcal{Q}_5$ töltésben, míg a diag-gauge klasszifikációban egy
$$\tilde{h}_j = \sum_{a,c=0}^{2} \gamma_{ac} P_j^a P_{j+1}^c \tag{1}$$
diagonális tag adódik ehhez.

2. Megoldások felcímkézése: szerepel a megoldás száma (no. n), a család neve (ez esetben Z3_RR, valamint a $\tilde{h}$ fajtája), a minta, vagyis a nem diagonális h^{ab} blokkok labeljei, a chart száma az adott mintában (ez a konstans szorzó, amit említettem múltkor), valamint az adott mintán belül a megtalált megoldás száma.

Mind a 9 blokk listázva, címkézve van, megfelelnek az ansatz formájának. A diagonális gauge esetén a 9 γ_{ac} paraméter van listázva, a hozzájuk tartozó operátorok csak projektorok, ugyebár.

1 KP_real

Genuine Solution no. 1 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 0:

$$h_1^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_1^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_1^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_1^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_1^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_1^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_1^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_1^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_1^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 3 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 2:

$$h_3^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_3^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_3^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_3^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_3^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_3^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_3^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_3^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_3^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 4 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 3:

$$h_4^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_4^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_4^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_4^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_4^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_4^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_4^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_4^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_4^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 5 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 4:

$$h_5^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_5^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_5^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_5^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_5^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_5^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_5^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_5^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_5^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 6 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 5:

$$h_6^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_6^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_6^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_6^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_6^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_6^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_6^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_6^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_6^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 7 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 6:

$$h_7^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_7^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_7^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_7^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_7^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_7^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_7^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_7^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_7^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

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Genuine Solution no. 8 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 7:

$$h_8^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_8^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_8^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_8^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_8^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_8^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_8^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_8^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_8^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 9 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 8:

$$h_9^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_9^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_9^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_9^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_9^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_9^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_9^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_9^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_9^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 10 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 9:

$$h_{10}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{10}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{10}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{10}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{10}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{10}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{10}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{10}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{10}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 11 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 10:

$$h_{11}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{11}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{11}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{11}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{11}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{11}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{11}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{11}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{11}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 12 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 11:

$$h_{12}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{12}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{12}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{12}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{12}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{12}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{12}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{12}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{12}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 13 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 12:

$$h_{13}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{13}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{13}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{13}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{13}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{13}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{13}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{13}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{13}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 14 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 13:

$$h_{14}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{14}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{14}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{14}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{14}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{14}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{14}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{14}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{14}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 15 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 14:

$$h_{15}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{15}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{15}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{15}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{15}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{15}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{15}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{15}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{15}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 16 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 15:

$$h_{16}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{16}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{16}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{16}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{16}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{16}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{16}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{16}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{16}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 17 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 16:

$$h_{17}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{17}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{17}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{17}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{17}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{17}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{17}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{17}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{17}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 18 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 17:

$$h_{18}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{18}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{18}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{18}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{18}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{18}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{18}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{18}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{18}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 19 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 18:

$$h_{19}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{19}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{19}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{19}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{19}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{19}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{19}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{19}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{19}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 20 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 19:

$$h_{20}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{20}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{20}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{20}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{20}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{20}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{20}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{20}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{20}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 21 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 20:

$$h_{21}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{21}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{21}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{21}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{21}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{21}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{21}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{21}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{21}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 22 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 21:

$$h_{22}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{22}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{22}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{22}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{22}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{22}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{22}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{22}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{22}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 23 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 22:

$$h_{23}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{23}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{23}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{23}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{23}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{23}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{23}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{23}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{23}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 24 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 23:

$$h_{24}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{24}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{24}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{24}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{24}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{24}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{24}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{24}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{24}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 25 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 24:

$$h_{25}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{25}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{25}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{25}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{25}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{25}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{25}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{25}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{25}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 26 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 25:

$$h_{26}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{26}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{26}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{26}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{26}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{26}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{26}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{26}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{26}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 27 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 26:

$$h_{27}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{27}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{27}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{27}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{27}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{27}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{27}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{27}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{27}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 28 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 27:

$$h_{28}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{28}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{28}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{28}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{28}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{28}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{28}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{28}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{28}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 29 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 28:

$$h_{29}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{29}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{29}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{29}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{29}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{29}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{29}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{29}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{29}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 30 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 29:

$$h_{30}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{30}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{30}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{30}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{30}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{30}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{30}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{30}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{30}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 31 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 30:

$$h_{31}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{31}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{31}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{31}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{31}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{31}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{31}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{31}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{31}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 32 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 31:

$$h_{32}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} - \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & u_{22}^{02} + \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{32}^{01} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 1 & 0 \\ 1 & u_{11}^{02} - u_{22}^{21} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{21} \end{pmatrix}$$

$$h_{32}^{02} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{32}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{32}^{11} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{32}^{12} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & -\frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{32}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & \frac{u_{22}^{21}}{2} \end{pmatrix}$$

$$h_{32}^{21} = \begin{pmatrix} -u_{11}^{02} + \frac{3u_{22}^{21}}{2} & 0 & 0 \\ 0 & -\frac{u_{22}^{21}}{2} & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{32}^{22} = \begin{pmatrix} -u_{22}^{02} + \frac{u_{22}^{21}}{2} & 0 & 0 \\ 0 & \frac{u_{22}^{21}}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 33 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 32:

$$h_{33}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{33}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{33}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{33}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{33}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{33}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{33}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{33}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{33}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 34 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 33:

$$h_{34}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{34}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{34}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{34}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{34}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{34}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{34}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{34}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{34}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 35 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 34:

$$h_{35}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{35}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{35}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{35}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{35}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{35}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{35}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{35}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{35}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 36 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 35:

$$h_{36}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{36}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{36}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{36}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{36}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{36}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{36}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{36}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{36}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 37 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 36:

$$h_{37}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{37}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{37}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{37}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{37}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{37}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{37}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{37}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{37}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 38 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 37:

$$h_{38}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{38}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{38}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{38}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{38}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{38}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{38}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{38}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{38}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 39 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 38:

$$h_{39}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{39}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{39}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{39}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{39}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{39}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{39}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{39}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{39}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 40 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 39:

$$h_{40}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{40}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{40}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{40}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{40}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{40}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{40}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{40}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{40}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 41 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 40:

$$h_{41}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{41}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{41}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{41}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{41}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{41}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{41}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{41}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{41}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 42 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 41:

$$h_{42}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{42}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{42}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{42}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{42}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{42}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{42}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{42}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{42}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 43 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 42:

$$h_{43}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{43}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{43}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{43}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{43}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{43}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{43}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{43}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{43}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 44 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 43:

$$h_{44}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{44}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{44}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{44}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{44}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{44}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{44}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{44}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{44}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 45 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 44:

$$h_{45}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{45}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{45}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{45}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{45}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{45}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{45}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{45}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{45}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 46 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 45:

$$h_{46}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{46}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{46}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{46}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{46}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{46}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{46}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{46}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{46}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 47 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 46:

$$h_{47}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{47}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{47}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{47}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{47}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{47}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{47}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{47}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{47}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 48 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 47:

$$h_{48}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{48}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{48}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{48}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{48}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{48}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{48}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{48}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & u_{22}^{21} \end{pmatrix}$$

$$h_{48}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 49 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 48:

$$h_{49}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{49}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{49}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{49}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{49}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{49}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{49}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{49}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{49}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 50 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 49:

$$h_{50}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{50}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{50}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{50}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{50}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{50}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{50}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{50}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{50}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 51 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 50:

$$h_{51}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{51}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{51}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{51}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{51}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{51}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{51}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{51}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{51}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 52 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 51:

$$h_{52}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{52}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{52}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{52}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{52}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{52}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{52}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{52}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{52}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 53 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 52:

$$h_{53}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{53}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{53}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{53}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{53}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{53}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{53}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{53}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{53}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 54 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 53:

$$h_{54}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{54}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{54}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{54}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{54}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{54}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{54}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{54}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{54}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 55 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 54:

$$h_{55}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{55}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{55}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{55}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{55}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{55}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{55}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{55}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{55}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 56 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 55:

$$h_{56}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{56}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{56}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{56}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{56}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{56}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$h_{56}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{56}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{56}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 57 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 56:

$$h_{57}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{57}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{57}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{57}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{57}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{57}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{57}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{57}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{57}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 58 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 57:

$$h_{58}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{58}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{58}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{58}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{58}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{58}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{58}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{58}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{58}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 59 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 58:

$$h_{59}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{59}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{59}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{59}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{59}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{59}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{59}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{59}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{59}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 60 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 59:

$$h_{60}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{60}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{60}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{60}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{60}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{60}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{60}^{20} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{60}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{60}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 61 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 60:

$$h_{61}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{61}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{61}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{61}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{61}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{61}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{61}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{61}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{61}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 62 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 61:

$$h_{62}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{62}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{62}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & -1 \\ 0 & u_{11}^{02} & 0 \\ -1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{62}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{62}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{62}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{62}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{62}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{62}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 63 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 62:

$$h_{63}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{63}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{63}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{63}^{10} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{63}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{63}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{63}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{63}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{63}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Genuine Solution no. 64 for family KP_real_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=0, comp 63:

$$h_{64}^{00} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{64}^{01} = \begin{pmatrix} -u_{11}^{02} + u_{22}^{21} & 1 & 0 \\ 1 & u_{11}^{02} & 0 \\ 0 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{64}^{02} = \begin{pmatrix} -u_{22}^{02} + u_{22}^{21} & 0 & 1 \\ 0 & u_{11}^{02} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_{64}^{10} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & u_{22}^{21} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{64}^{11} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$h_{64}^{12} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & u_{22}^{21} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$h_{64}^{20} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & u_{22}^{21} \end{pmatrix}$$

$$h_{64}^{21} = \begin{pmatrix} -u_{11}^{02} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & u_{22}^{21} \end{pmatrix}$$

$$h_{64}^{22} = \begin{pmatrix} -u_{22}^{02} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$